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ULTRASONIC CATHETER

Abstract

An ultrasonic catheter is disclosed. The ultrasonic catheter comprises a micro-electromechanical (MEMS) based Piezoelectric Micromachined Ultrasonic Transducer (pMUT) array disposed within a distal end of a body. The ultrasonic catheter further comprises a plurality of electronic flex circuits, connected at one end to a handle connector and at other end to a first layer of AIC. The handle connector passes ultrasound signals to the distal end via the plurality of electronic flex circuits. Further, a distal pMUT interposer is disposed between the first layer and a second layer of AIC, with Invisi Via connections. The Invisi Via connection is configured to bring the electronic flex circuit from one side of the distal pMUT interposer to the other. Further, the MEMS based pMUT array sends a pressure wave and receives a returning pressure wave echo via the InvisiVia connections of the distal pMUT interposer.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/052,926 entitled “ULTRASONIC CATHETER,” filed Nov. 6, 2022, which is hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates generally to the field of ultrasonic catheter. More particularly, some embodiments relate to ultrasonic catheters having a distal piezoelectric micromachined transducer (pMUT) interposer referred as “InvisiVia” connections, for transmitting and receiving acoustic pulse information.

BACKGROUND OF THE DISCLOSURE

[0003] Use of catheter-based structural and electrophysiological procedures have recently expanded to more complex scenarios, in which an accurate definition of variable individual cardiac anatomy is a key to obtain optimal results. Intracardiac echocardiography (ICE) is a unique imaging modality for high-resolution real-time visualization of cardiac structures, continuous monitoring of catheter location within the heart, and early recognition of procedural complications, such as pericardial effusion or thrombus formation. Further, ICE imaging modality includes additional benefits, such as excellent patient tolerance, reduction of fluoroscopy time, and elimination of need for general anaesthesia or second operator. Currently, ICE imaging modality has largely replaced trans-oesophageal echocardiography as ideal imaging modality for guiding certain procedures, such as atrial septal defect closure and catheter ablation of cardiac arrhythmias, and has an emerging role in others, including mitral valvuloplasty, transcatheter aortic valve replacement, and left atrial appendage closure.

[0004] ICE catheters involve interconnection of multidimensional transducer arrays with electronics. A technological challenge for the multidimensional transducer arrays is to create a secure and stable interconnection between an acoustic array and associated echoes. There are hundreds of different elements distributed in two dimensions which require interconnection along z-axis (in depth or range) for some elements surrounded by other elements. Since the elements are small in size, there is a limited space for a separate electrical connection to each element. There are a variety of approaches to provide interconnection for the multidimensional transducer array, such as, a chip-on-array, a frame-based approach, and a multi-layered flex. In the chip-on-array, the acoustic array is built-up directly on an application specific integrated circuit (ASIC) chip's input/output (I/O). Due to possible failure in the acoustic array a very low process yield may result. [0005] In the frame-based approach, the array is divided into few sections. Each section has a solid metal frame. Flex circuits bent around the metal frames redistribute as many signals as a number of acoustic elements from the acoustic elements to ASICs placed on the flex circuits. However, signal routing of the frame-based approach is more complicated than chip-on-array, which results in higher electrical parasitic. In the multi-layered flex approach, stacks may be divided into flex circuits to route the signals for elements without any aperture sectioning. However, these approaches either end up in an expensive structure of interconnection for the multidimensional transducer array, or there is a certain undesired leakage to bulk section substrate of the transducer array.

[0006] Therefore, there is a need for an improved ultrasonic catheter with a high-density flexible circuit having electrical insulated interconnects to enable highly repeatable and stable transmission of signals.

SUMMARY OF THE DISCLOSURE

[0007] By way of introduction, the preferred embodiments described below include an easy-to-use ultrasonic catheter is disclosed. The ultrasonic catheter comprises a body having a longitudinal axis, a proximal end, and a distal end. Further, the ultrasonic catheter comprises a micro-electromechanical (MEMS) based Piezoelectric Micromachined Ultrasonic Transducer (pMUT) array disposed within the distal end of the body. The ultrasonic catheter further comprises a plurality of electronic flex circuits, connected at one end to a handle connector and at other end to a first layer of Adhesive Interconnect (AIC). The handle connector passes ultrasound signals to the distal end via the plurality of electronic flex circuits. Further, the ultrasonic catheter comprises a distal pMUT interposer disposed between the Flex circuit layer and a second layer of Anisotropic Conductive Film (ACF) or similar AIC, with what we refer to as “Invisi Via” connections.

[0008] The Invisi Via feature provides a conductive interconnect from one side of the flexible circuit to the other side of the flexible circuit. Thus, the InvisiVia creates a flat and planar interconnect from the flex circuitry to the flex bonding pads by way of a thermocompression electrical connection. Further, the MEMS based pMUT array receives an electronic pulse through the flex circuit and through the conductive Invisi Via to the pMUT, generating a pressure wave and the returning pressure wave stimulates the pMUT creating a return an electrical signal through the conductive Invisi Via connections of the distal pMUT interposer.

[0009] It can be noted that the distal pMUT interposer corresponds to an ultra-high-density interconnect (UHDI) flexible circuit interposer connected to the MEMS based pMUT array through an UHDI pad array. The UHDI pad array is connected to the MEMS based pMUT array through a use of near zero evidence interfacial via. Further, the use of conductive Invisi Via connections creates a higher percentage of flat and planar conductive surface area on the UHDI pad. In one embodiment, the Invisi Via connections creates a flat and planner interconnect diameter of less than 15 microns. Thus, the use of Invisi Via connections eliminates a potential for undesired surface topography, creates a larger percentage of conductive surface are and helps prevent leakage to bulk silicon substrate of the MEMS based pMUT array.

[0010] Other features and aspects of this disclosure will be apparent from the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Various embodiments will hereinafter be described in accordance with the appended drawings, which are provided to illustrate and not to limit the scope of the disclosure in any manner, wherein similar designations denote similar elements, and in which:

[0012] FIG. 1 illustrates a perspective view of an ultrasonic catheter, according to an embodiment of the present disclosure;

[0013] FIG. 2 illustrates a schematic diagram of a front view of the ultrasonic catheter, according to an embodiment of the present disclosure;

[0014] FIG. 3 illustrates an interconnect diagram showing a side view stack-up of electronic flex connection to a micro-electromechanical (MEMS) based Piezoelectric Micromachined Ultrasonic Transducer (pMUT) array or other type of Transducer, through a layer or layers of Adhesive Interconnect (AIC) such as Anisotropic Conductor Film (ACF), Anisotropic Conductive Paste (ACP), or similar An isotropic Conductive Adhesives (ACAs) or through Electrically Conductive Adhesives (EC As) or other such conductive adhesive, according to an embodiment of the present disclosure;

[0015] FIG. 4A illustrates a side view of InvisiVia or blind via connection shape, according to an embodiment of the present disclosure;

[0016] FIG. 4B illustrates the Invisi Via connection with a flat and planar interconnect from a flex

circuitry to a highly planar attachment land/pad surface, according to an embodiment of the present disclosure;

[0017] FIG. 4C illustrates a three-dimensional view with the Invisi Via connection disposed within the distal pMUT interposer, according to an embodiment of the present disclosure;

[0018] FIG. 5 illustrates a schematic diagram showing a first layer of AIC, connecting a plurality of electronic flex circuits to a distal pMUT interposer, according to an embodiment of the present disclosure;

[0019] FIG. 6. illustrates an interconnect diagram showing a top view stackup of the Invisi Via connection to the MEMS based pMUT array through a second layer of AIC, according to an embodiment of the present disclosure;

[0020] FIG. 7 illustrates a schematic diagram showing the AIC layer connecting the distal pMUT interposer to the MEMS based pMUT, according to an embodiment of the present disclosure; and

[0021] FIGS. 8-9 illustrate cross sectional images of a heart with the ultrasonic catheter positioned within a right atrium of the heart, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0022] The components of the embodiments as generally described and illustrated in the figures herein can be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of various embodiments, as represented in the figures, is not intended to limit the scope of the present disclosure but is merely representative of various embodiments. While various aspects of the embodiments are presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

[0023] Some embodiments of this disclosure, illustrating all its features, will now be discussed in detail. The words “comprising,” “having,” “containing,” and “including,” and other forms thereof, are intended to be equivalent in meaning and be open-ended in that an item or items following any one of these words is not meant to be an exhaustive listing of such item or items, or meant to be limited to only the listed item or items.

[0024] It must also be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural references unless the context dictates otherwise. Although any systems and methods similar or equivalent to those described herein can be used in the practice or testing of embodiments of the present disclosure, the preferred systems, and methods are now described. The terms “proximal” and “distal” are opposite directional terms. For example, the distal end of a device or component is the end of the component that is furthest from the practitioner during ordinary use. The proximal end refers to the opposite end, or the end nearest the practitioner during ordinary use.

[0025] Embodiments of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings in which like numerals represent like elements throughout the several figures, and in which example embodiments are shown. Embodiments of the present disclosure may, however, be embodied in alternative forms and should not be construed as being limited to the embodiments set forth herein. The examples set forth herein are non-limiting examples and are merely examples among other possible examples.

[0026] FIG. 1 illustrates a perspective view of an ultrasonic catheter **100**, according to an embodiment of the present disclosure. The ultrasonic catheter **100** may comprise a body **102** having a longitudinal axis **104**, a proximal end **106**, and a distal end **108**, a handle assembly **110**, a steering control unit **112** having a steering handle **114** and a housing **116**, and a distal tip **118**.

[0027] The handle assembly **110** may be positioned between the proximal end **106** and the distal end **108** of the ultrasonic catheter **100**. Further, the steering control unit **112** may be positioned within the handle assembly **110**. The steering control unit **112** may be provided for articulating the distal tip **118** of the ultrasonic catheter **100**. Further, the steering control unit **112** may align face of a micro-electromechanical (MEMS) based Piezoelectric Micromachined Ultrasonic Transducer (pMUT) array (not shown) towards different view including an anterior position and a posterior

position inside the heart. Further, the steering control unit **112** may comprise the steering handle **114** and the housing **116** enclosing an actuator (not shown) and a steering hub (not shown). It can be noted that an internal friction occurs between the actuator and the steering hub, and between the actuator and the housing **116**, which causes the ultrasonic catheter **100** to retain its adjusted configuration without operator attention. Further, the steering handle **114** may be rotated to facilitate positioning of the distal tip **118** of the ultrasonic catheter **100**.

[0028] In one embodiment, the steering handle **114** may be rotated to position the distal tip **118** inside a chamber of a heart of a patient. In one embodiment, the steering control unit **112** may comprise a set of steering lines controlled by the steering control unit **112** to articulate multi-directionally a distal segment of the ultrasonic catheter **100** when placed inside the heart. Further, the ultrasonic catheter **100** may be disposed within the chamber of the heart of a patient and coupled to an imaging system (not shown) using a dongle cable (not shown) for displaying two-dimensional (2D) or three-dimensional (3D) images of the chamber of the heart using ultrasound waves and acoustic pulses.

[0029] Further, the ultrasonic catheter **100** may be used to perform electrophysiology (EP). The ultrasonic catheter **100** may be used for diagnosis and/or treatment in combination with another imaging modality, such as an x-ray, fluoroscopy, magnetic resonance, computed tomography, or optical system. Both imaging modalities may scan a patient for generating images to assist a physician. The data from the different modalities may be aligned by locating markers with a known spatial relationship to the ultrasound scan in the images of the other modality. In other embodiments, the ultrasonic catheter **100** may be a flexible cylindrical section without the markers and/or without another imaging modality. In one embodiment, the ultrasonic catheter **100** may utilize a microelectromechanical (MEMS) system defined as piezoelectric micro-machined ultrasound transducer (pMUT) or other types of MEMS transducers, interconnected using matched flexible circuits. In one embodiment, the ultrasonic catheter **100** may correspond to an intracardiac echocardiography (ICE) MEMS ultrasonic catheter utilizing a high-density flexible circuit for all transmission and electrical interconnects. It can be noted that the use of the high-density flexible circuits may enable highly repeatable and stable transmission and return signals. Further, the high-density flexible circuit transmission lines may transmit electrical energy from one end to another end of the ultrasonic catheter **100**.

[0030] Further, the ultrasonic catheter **100** may comprise a catheter shaft **120**. The catheter shaft **120** may be coupled to the handle assembly **110** at one end and to the distal tip **118** of the ultrasonic catheter **100** at the other end. Further, the catheter shaft **120** may enclose an electronic flexible circuit (not shown) and a plurality of steering cables (not shown). In one embodiment, the electronic flex circuit may be referred to as flexible cables. It can be noted that the electronic flex circuit may be bend or tilt towards the anterior position and/or posterior position inside the chamber of the heart. In one embodiment, the electronic flex circuit may comprise a strand, wire, and/or thread, and is preferably made from a low profile, durable, nonelastic, and non-conducting material. In one embodiment, steering cables or steering wires may be made from stainless steel. In another embodiment, the steering cables may be made of synthetic materials, such as nylon or similar synthetic fibres, or plastics material, such as urethane, Teflon®, Kynar®, Kevlar®, polyethylene, multi-stranded nylon, or gel-spun polyethylene fibres. For example, the steering cables may be a multi-stranded Spectra® brand nylon line sold as Spiderwire® fishing line (10 lbs. test).

[0031] The ultrasonic catheter **100** may be provided to transmit ultrasound signals inside the chamber of the heart of the patient. In one embodiment, the ultrasonic catheter **100** may be a flexible elongate member with the body **102** having the catheter shaft **120** at one end and the handle assembly **110** at another end. In one embodiment, the distal tip **118** of the ultrasonic catheter **100** may be coated with an insulating material. Further, the insulating material may be disposed at the distal end **108** over an imaging window (not shown), to prevent electrical breakdown and

leakage of electrical signals from the ultrasonic catheter **100**. The insulating material may be a copolymer material such as, but not limited to, polyether block amide (PEBA, for example, available under the trade name PEBAXX), and a thermoplastic elastomer (TPE). In one embodiment, the insulating material provides a low loss acoustic window. It can be noted that the insulating material provides a low loss path of acoustic echo when the ultrasonic catheter **100** is inserted inside the heart of the patient.

[0032] FIG. **2** illustrates a schematic diagram of the ultrasonic catheter **100**, according to an embodiment of the present disclosure. FIG. **2** is described in conjunction with FIG. **1**.

[0033] The ultrasonic catheter **100** may comprise a micro-electromechanical (MEMS) based Piezoelectric Micromachined Ultrasonic Transducer (pMUT) array **202** and a substrate **204**. The MEMS based pMUT array **202** may be disposed over the substrate **204** towards the distal end **108** of the ultrasonic catheter **100**. Further, the MEMS based pMUT array **202** may be disposed within the distal tip **118** of the ultrasonic catheter **100**.

[0034] The catheter shaft **120** may be coupled between the handle assembly **110** via a distal pMUT interposer (not shown) and the MEMS based pMUT array **202**. The electronic flex circuit inside the catheter shaft **120** may receive the at least one signal from the MEMS based pMUT array **202** and the received signal may be communicated back to an imaging device (not shown). Further, the MEMS based pMUT array **202** may send pressure waves to a target and receive a returning pressure wave echo from the target. In one embodiment, the target may be inside the chamber of heart of the patient. The MEMS based pMUT array **202** may transmit the received pressure wave echo back to the imaging device via the electronic flex circuit to further analyze the pressure wave echo for image generation. The MEMS based pMUT array **202** comprises a plurality of MEMS based pMUT array elements (not shown) arranged on the substrate **204**.

[0035] In one embodiment, the ultrasonic catheter **100** may employ MEMS based pMUT array **202** with bulk piezoelectric transducer (PZT), piezoelectric micro machined (PMUT) transducer or capacitor micro machined transducer (CMUT) array for introduction through subclavian access for lead implantation for pacemaker, defibrillator, and structural heart implantation and other cardiac procedures.

[0036] FIG. **3** illustrates an interconnect diagram showing a side view stackup of electronic flex connection to a micro-electromechanical (MEMS) based Piezoelectric Micromachined Ultrasonic Transducer (pMUT) array, through layers of AIC, according to an embodiment of the present disclosure. FIG. **3** is described in conjunction with FIGS. **1** and **2**.

[0037] Referring to FIG. **3**, a compression connection is made among a catheter handle connector **302**, a plurality of electronic flex circuits **304**, the MEMS based pMUT array **202**, and a distal pMUT interposer **306**, using a first layer of AIC **308** and a second layer of AIC **310**. The AIC may be made of a material selected from a group of materials of Anisotropic Conductor Film (ACF), anisotropic conductive paste (ACP), or similar anisotropic conductive adhesives (ACAs) or through Electrically Conductive Adhesives (ECAs) or other such conductive adhesive. The first layer of AIC **308** and the second layer of AIC **310** may sandwich the distal pMUT interposer **306** and create the compression connection between the plurality of electronic flex circuits **304** and the distal pMUT interposer **306**. Further, the first layer of AIC **308** may be coupled to the catheter handle connector **302** via the plurality of electronic flex circuits **304**, and the second layer of AIC **310** may be coupled between the MEMS based pMUT array **202** and the distal pMUT interposer **306**. Further, the catheter handle connector **302** may pass ultrasound signals through the plurality of electronic flex circuits **304** to the distal end **108** of the ultrasonic catheter **100**. Further, the catheter handle connector **302** may comprise board edge connectors (not shown) and connector pins (not shown) to pass the ultrasound signals towards the distal end **108** of the ultrasonic catheter **100**.

[0038] In one embodiment, the first layer of AIC **308** and the second layer of AIC **310** may provide a lead-free and environment friendly adhesive interconnect system. The first layer of AIC **308** and the second layer of AIC **310** may be used to create an electrically conductive adhesive bond for

flexible and rigid circuits, as shown in FIG. 4A. Further, the first layer of AIC 308 and the second layer of AIC 310 may provide an extremely fine pitch capability for the pressure echo received from the MEMS based pMUT array 202.

[0039] Further, the distal pMUT interposer 306 sandwiched between the first layer of AIC 308 and the second layer of AIC 310, with Invisi Via connections 402, as shown in FIG. 4A. In one embodiment, the distal pMUT interposer 306 may be referred to as a distal PMUT flex cable interposer. Further, the distal pMUT interposer 306 may correspond to an ultra-high density (UHD) flexible circuit interposer connected to the MEMS based pMUT array 202 through a UHD interconnect pad (not shown). The UHD interconnect pad may be connected to the MEMS based pMUT array 202 through a use of near zero evidence interfacial via. In one embodiment, the first layer of the AIC 308 may provide connections between the plurality of electronic flex circuits 304 and the UHD interconnect pad. The second layer of the AIC 310 may provide connections between the UHD interconnect pad, and the MEMS based pMUT array 202.

[0040] It should be noted that the MEMS based pMUT array 202 may send a pressure wave and receive a returning pressure wave echo via the InvisiVia connections 402 of the distal pMUT interposer 306. Further, the use of the Invisi Via connections 402 may create a flat and planar interconnect with the UHD interconnect pad. In an example, the Invisi Via connections 402 creates a flat and planner interconnect diameter of not more than 5 μm . Further, the Invisi Via connections 402 may be highly critical to an electrical performance of the ultrasonic catheter 100. In one embodiment, the Invisi Via connections 402 may eliminate a potential for undesired leakage to a bulk silicon substrate of the MEMS based pMUT array 202. In one embodiment, the Invisi Via connections 402 may be from 16 to 128.

[0041] Referring to FIG. 4B, the Invisi Via connection 402 with a flat and planar interconnect from a flex circuitry to a highly planer attachment land/pad surface, is disclosed, according to an embodiment of the present disclosure.

[0042] The InvisiVia connection 402 may be configured to bring the electronic flex circuit connections from one side of the distal pMUT interposer 306 to the other side. It can be noted that purpose of the Invisi Via connection 402 is that the second layer of AIC 310 should not connect to silicon disposed on side of the MEMS based pMUT array 202. Further, the Invisi Via connection 402 may create a flat and planar interconnect from the electronic flex circuit to second layer of AIC 310, as shown in FIGS. 5 and 6. In one embodiment, the flex bonding pads are electrically connected to the MEMS based pMUT array 202 utilizing a thermal compression adhesive system.

[0043] Further the Invisi Via connection 402 may be a conical section having a first end 404 and a second end. The Invisi Via connection 402 may be disposed within the distal pMUT interposer 306, as shown in FIG. 4C. The first end 404 and the second end 406 may have diameter ranging from 10 μm to 25 μm and Oum to Bum respectively. In one embodiment, the Invisi Via connection 402 is provided with an Invisi Via plating 408 between the first end 404 and the second end 406. In one embodiment, the first end 404 and the second end 406 may correspond to a top and bottom end of the Invisi Via connection 402. The first end 404 of the Invisi Via connection 402 may be connected to the second layer of AIC 310 via a trace 412. Further, the second end 406 of the Invisi Via connection 402 may connect through the second layer of AIC 310 to the MEMS based pMUT array 202. It can be noted that the Invisi Via connection 402 may facilitate the MEMS based pMUT array 202 with an extremely flat and planar surface with maximum surface area due to no or minimal copper penetration.

[0044] In one embodiment, the distal pMUT interposer 306 is aligned and bonded, and the pressure echo may be transmitted electrically through the InvisiVia connections 402 of the distal pMUT interposer 306. This provides for z-axis routing of the ultrasound signals from the MEMS based pMUT array 202. The first layer of AIC 308 and the second layer of AIC 310 may include a bonding material to physically hold the distal pMUT interposer 306 to the MEMS based pMUT array 202. The Invisi Via connections 402 may be created from a z-axis interconnect of two or

more signals or ground layers utilizing an ultraviolet (UV) laser **410** with a hole diameter of no more than 10 μm that is subsequently copper plating to a closed and highly planar attachment land/pad surface. In one embodiment, the UV laser **410** may have a beam width of less than 25 μm . [0045] FIG. **5** illustrates a schematic diagram showing the first layer of AIC **308**, connecting the plurality of electronic flex circuits **304** to the distal pMUT interposer **306**, according to an embodiment of the present disclosure.

[0046] Further, the distal pMUT interposer **306** may be coupled to the plurality of electronic flex circuits **304** via the first layer of AIC **308**. The first layer of AIC **308** may comprise a plurality of conductive particles **502**. It can be noted that the plurality of conductive particles **502** may create the compression connection between the plurality of electronic flex circuits **304** and the distal pMUT interposer **306**. Therefore, a stable transmission of ultrasound signals is achieved by the first layer of AIC **308** between the plurality of electronic flex circuits **304** and the distal pMUT interposer **306**. The first layer of AIC **308** may provide an extremely fine pitch capability for the pressure echo received from the MEMS based pMUT array **202**.

[0047] FIG. **6** illustrates an interconnect diagram showing a top view stackup of the Invisi Via connection **402** to the MEMS based pMUT array **202** through the second layer of AIC **310**, according to an embodiment of the present disclosure.

[0048] Further, the distal pMUT interposer **306** may be coupled to the MEMS based pMUT array **202** via the second layer of AIC **310**, as shown in FIG. **6**. It can be noted that the distal pMUT interposer **306** may create the compression connection between the Invisi Via connections **402** of the distal pMUT interposer **306** and the MEMS based pMUT array **202** via the plurality of conductive particle **502** of the second layer of AIC **310**. It can also be noted that the second layer of AIC **310** may also provide the extremely fine pitch capability for the pressure echo received from the MEMS based pMUT array **202**. In one embodiment, the Invisi Via connections **402** may create a flat and planner interconnect diameter of not more than Sum.

[0049] FIG. **7** illustrates a schematic diagram showing the AIC layer connecting the distal pMUT interposer **306** to the MEMS based pMUT array **202**, according to an embodiment of the present disclosure.

[0050] The plurality of electronic flex circuits **304** is connected to the distal pMUT interposer **306** via the first layer of AIC **308**. The distal pMUT interposer **306** is connected to the MEMS based pMUT array **202** via the second layer of AIC **310**. It can be noted that the distal pMUT interposer **306** may correspond to the UHD flexible circuit interposer connected to the MEMS based pMUT array **202** through an UHD interconnect pad. The UHD interconnect pad may be connected to the MEMS based the pMUT array **202** through a use of near zero evidence interfacial via. Further, the use of InvisiVia connections **402** may create the flat and planner interconnect with the MEMS based pMUT array **202**. Further, the compression connections may be made between the plurality of electronic flex circuits **304**, the first layer of AIC **308**, the distal pMUT interposer **306**, the second layer of AIC **310**, and the MEMS based pMUT array **202**. In one embodiment, the use of Invisi Via connections **402** may eliminate a potential for undesired leakage to bulk silicon substrate of the MEMS based pMUT array **202**.

[0051] Referring to FIGS. **8-9**, cross sectional images of a heart **800** with the ultrasonic catheter **100** positioned within a right atrium **802** of the heart **800**, are disclosed. The distal tip **118** of the ultrasonic catheter **100** may be inserted into the right atrium **802** via an inferior vena cava (not shown). In order to perform an adequate imaging of an interatrial septum (IAS) **804** and its neighbouring structures, two standardized views may be used. The movement of the distal tip **118** of the ultrasonic catheter **100** within the right atrium **802** may be controlled by the steering control unit **112**. Further, the movement of the distal tip **118** by the steering control unit **112** clockwise or counterclockwise, may allow the imaging window to move from posterior view to anterior view or vice versa.

[0052] Further, as shown FIGS. **8-9**, to properly place the MEMS based pMUT array **202** into

position for imaging the right atrium **802** and a bicuspid valve **806**, the flexible sheath may be introduced into the patient's vascular structure via a femoral vein (not shown). Using a fluoroscopic imaging to monitor the ultrasonic catheter's position, a clinician may advance the distal end **108** of the ultrasonic catheter **100** into the right atrium **802**. In order to guide the ultrasonic catheter **100** through turns in the patient's vascular structure, the clinician may rotate the distal tip **118** clockwise or counterclockwise to allow the imaging window to move towards from the anterior position to the posterior position and vice versa. Once the distal tip **118** of the ultrasonic catheter **100** is in the right atrium **802**, the clinician may rotate the distal tip **118** so as to introduce an acute bend in the flexible sheath to direct the MEMS based pMUT array **202** through a tricuspid valve **808** and into a right ventricle **810**, as shown in FIG. **6**. In this position, a field of view of the MEMS based pMUT array **202** may include portions of the right ventricle **810**, the IAS **804**, the bicuspid valve **806**, a left ventricle **812**, a left atrium **814** and a left ventricular wall. In one embodiment, when the MEMS based pMUT array **202** is directed clockwise, the right ventricle **810** and right ventricular wall may be imaged. It can be noted that the clinician may twist the ultrasonic catheter **100**, while positioned within the heart **800** as illustrated in FIGS. **8-9**, the MEMS based pMUT array **202** may swing about an axis, which could injure the tricuspid valve **808** or cause the MEMS based pMUT array **202** to strike the IAS **804**. FIG. **8** shows a perpendicular short-axis view to visualize interior parts of the IAS **804**. The interior parts of the IAS **804** include an aorta **902**, a mitral valve **904** towards the left ventricle **812**.

[0053] In one exemplary embodiment, the standard view is obtained by placing the ultrasonic catheter **100** in a mid-right atrium and the MEMS based pMUT array **202** in a neutral position facing the tricuspid valve **808**. The standard view provides imaging of the right atrium **802**, the tricuspid valve **808**, the right ventricle **810**, and typically an oblique or short-axis view of the aortic valve.

[0054] Further, when the ultrasonic catheter **100** may be rotated clockwise the aortic valve in long axis and the right ventricle **810** outflow tract is viewed. In this view, the tricuspid valve **808** that is closer to the MEMS based pMUT array **202** or the distal tip **118** is a non-coronary cusp, which is in close relationship to a membranous septum and a para-hisian region, whereas the opposite is a right coronary cusp, which is the most anterior of the aortic cusps, directly posterior to the right ventricle **810** outflow tract infundibulum and pulmonic valve. The left ventricle **812** is visualized anterior to the most septal portion of the right atrium **802**, and the opening of the coronary sinus becomes evident. In this view, the long axis of the left ventricle **812** outflow tract is identified, and the posterior left ventricle **812** is in view just below the non-coronary cusp.

[0055] Further, an additional clockwise rotation of the ultrasonic catheter **100** allows visualization of the mitral valve **904** and the IAS **804**, with the left atrial appendage anteriorly and the coronary sinus posteriorly. The left atrium **814** appendage is examined for the presence of thrombus at its ostium, and mitral regurgitation may be assessed using a colour Doppler.

[0056] In one embodiment, most catheters used in intravascular applications, particularly those with ultrasound transducers, are at least about 6 French in diameter. The electronics and wires needed for ultrasound transducer arrays have made it impractical and expensive to reduce the size of such catheters below about 6 French. Nevertheless, there are benefits in reducing the diameter of the catheter, and technology advances may enable the electronics and control structures to be further reduced in size. The bundling arrangement of the coaxial cables, steering and pivot cables and steering and pivot mechanisms described in more detail below, make it possible to effectively reduce the diameter below about 6 French, to about 5 French, about 4 French, about 3 French, or even about 2 French.

[0057] While there is shown and described herein certain specific structures embodying various embodiments of the invention, it will be manifest to those skilled in the art that various modifications and rearrangements of the parts may be made without departing from the spirit and

scope of the underlying inventive concept and that the same is not limited to the particular forms herein shown and described except insofar as indicated by the scope of the appended claims.

Claims

1. An ultrasonic catheter comprising: a catheter handle connector; a plurality of flex cables connected to the catheter handle connector; a first adhesive interconnect (AIC) connected to the plurality of flex cables; a distal piezoelectric micromachined ultrasonic transducer (pMUT) interposer connected to the first AIC; a second AIC connected to the distal pMUT interposer; and a pMUT array connected to the second AIC.
2. The ultrasonic catheter of claim 1, wherein the distal pMUT interposer includes Invisi Via connections configured to bring electronic signals from one side of the distal pMUT interposer to the other.
3. The ultrasonic catheter of claim 2, wherein the Invisi Via connections create a flat and planar interconnect diameter of not more than 5 μm .
4. The ultrasonic catheter of claim 2, wherein each of the Invisi Via connections is created from a z-axis interconnect of two or more signals or ground layers utilizing an ultraviolet (UV) laser with a hole diameter of no more than 10 μm that is subsequently copper plated to form a closed and highly planar attachment surface.
5. The ultrasonic catheter of claim 2, wherein each of the Invisi Via connections has a first end and a second end, with diameters ranging from 5 μm to 25 μm and 0 μm to 10 μm respectively.
6. The ultrasonic catheter of claim 1, wherein a compression connection is made among the plurality of flex cables, the first AIC, the distal pMUT interposer, the second AIC, and the pMUT array.
7. The ultrasonic catheter of claim 1, wherein the distal pMUT interposer comprises an ultra-high density (UHD) flexible circuit interposer connected to the pMUT array through an ultra-high density (UHD) interconnect pad.
8. The ultrasonic catheter of claim 7, wherein the UHD interconnect pad is connected to the pMUT array through a use of near zero evidence interfacial via.
9. The ultrasonic catheter of claim 1, further comprising a catheter body having a longitudinal axis, wherein the pMUT array is configured to send a pressure wave and receive a returning pressure wave echo through the distal pMUT interposer.
10. The ultrasonic catheter of claim 2, wherein the Invisi Via connections are between 30 to 128 in number.
11. The ultrasonic catheter of claim 1, wherein the pMUT array comprises a plurality of linear phased array elements each configured to create an individually focused beam.
12. An ultrasonic catheter, comprising: a catheter body having a proximal portion and a distal portion; a handle connector at the proximal portion of the catheter body; a micro-electromechanical (MEMS) based Piezoelectric Micromachined Ultrasonic Transducer (pMUT) array disposed within the distal portion of the catheter body, wherein the MEMS based pMUT array comprises a plurality of linear phased array elements each configured to create an individually focused beam; an ultra-high density (UHD) flexible circuit interposer connected to the MEMS based pMUT array through an ultra-high density (UHD) interconnect pad, wherein the UHD flexible circuit interposer includes near zero evidence interfacial vias connecting to the MEMS based pMUT array; a first layer of Adhesive Interconnect (AIC) positioned proximal to the UHD flexible circuit interposer; a second layer of Adhesive Interconnect (AIC) positioned distal to the UHD flexible circuit interposer; a plurality of Invisi Via connections formed through the UHD flexible circuit interposer; and a plurality of electronic flex circuits connecting the handle connector to the first layer of AIC.
13. The ultrasonic catheter of claim 12, wherein each of the Invisi Via connections has a first end with a diameter ranging from 5 μm to 25 μm and a second end with a diameter ranging from 0 μm

to 10 μm .

14. The ultrasonic catheter of claim 12, wherein: the first layer of AIC provides connections between the plurality of electronic flex circuits and the UHD interconnect pad; and the second layer of AIC provides connections between the UHD interconnect pad and the MEMS based pMUT array.

15. A method of manufacturing an ultrasonic catheter, comprising: providing a catheter body having a longitudinal axis, a proximal end, and a distal end; disposing a micro-electromechanical (MEMS) based Piezoelectric Micromachined Ultrasonic Transducer (pMUT) array within the distal end of the catheter body; forming a distal pMUT interposer having first and second sides; creating Invisi Via connections through the distal pMUT interposer by: forming holes through the distal pMUT interposer using an ultraviolet (UV) laser, wherein the holes have a diameter of no more than 10 μm , and copper plating the holes to form the Invisi Via connections with a planar attachment surface; positioning the distal pMUT interposer between a first layer of Adhesive Interconnect (AIC) and a second layer of AIC; connecting a plurality of electronic flex circuits between a handle connector and the first layer of AIC; and establishing a compression connection among the plurality of electronic flex circuits, the first layer of AIC, the distal pMUT interposer, the second layer of AIC, and the MEMS based pMUT array.

16. The method of manufacturing an ultrasonic catheter of claim 15, wherein the MEMS based pMUT array comprises a plurality of linear phased array elements, each configured to create an individually focused beam.

17. The method of manufacturing an ultrasonic catheter of claim 15, wherein the Invisi Via connections are between 30 to 128 in number, and wherein each of the Invisi Via connections provides an extremely flat and planar surface with maximum surface area due to minimal copper penetration.

18. The method of manufacturing an ultrasonic catheter of claim 15, wherein the distal pMUT interposer corresponds to an ultra-high density (UHD) flexible circuit interposer connected to the MEMS based pMUT array through an ultra-high density (UHD) interconnect pad, wherein the UHD interconnect pad is connected to the MEMS based pMUT array through a use of near zero evidence interfacial via.

19. The method of manufacturing an ultrasonic catheter of claim 15, wherein each of the Invisi Via connections has a first end and a second end, with diameters ranging from 5 μm to 25 μm and 0 μm to 10 μm respectively, and wherein each of the Invisi Via connections is provided with an Invisi Via plating between the first end and the second end.

20. The method of manufacturing an ultrasonic catheter of claim 15, wherein the first and second layers of AIC are made of a material selected from a group of materials consisting of Anisotropic Conductor Film (ACF), Anisotropic Conductive Paste (ACP), Anisotropic Conductive Adhesives (ACAs), and Electrically Conductive Adhesives (ECAs).
